

Master MVA - Cours de remise à niveau

14/09/26

Cours n°1 - Probabilités

I - Probability spaces

Ω : "univers"

Definition: σ -algebra (tribu en français)

a set $\mathcal{F} \in \mathcal{P}(\Omega)$ s.t.:

- ① $\emptyset \in \mathcal{F}$ ② $A \in \mathcal{F} \rightarrow A^c \in \mathcal{F}$ ③ $(\forall n \in \mathbb{N}, A_n \in \mathcal{F}) \Rightarrow \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{F}$

Example:

- $\{\emptyset, \Omega\}$
- $\mathcal{P}(\Omega)$
- $\{\emptyset, A, A^c, \Omega\}$ with $A \subset \Omega$

Proprieties:

- $\Omega \in \mathcal{F}$
- $\forall n, A_n \in \mathcal{F} \Rightarrow \bigcap A_n \in \mathcal{F}$

Demo: $\left(\left(\bigcap_{n \in \mathbb{N}} A_n \right)^c \right)^c = \underbrace{\left(\bigcup_{n \in \mathbb{N}} \underbrace{A_n^c}_{\substack{\in \mathcal{F} \\ \in \mathcal{F}}} \right)^c}_{\substack{\in \mathcal{F} \\ \in \mathcal{F}}} = \bigcap A_n \in \mathcal{F}$

Definition: a probability measure is an application $P: \mathcal{F} \rightarrow [0,1]$ s.t.

- $P(\Omega) = 1$
 - If $(A_n)_{n \in \mathbb{N}}$ are disjoint, $P(\bigcup A_n) = \sum P(A_n)$
- $(\Omega, \mathcal{F}, P)$: probability space

Example: Ω finite, $\mathcal{F} = \mathcal{P}(\Omega)$
 $P(A) = \frac{\text{card}(A)}{\text{card}(\Omega)}$

properties:

- $P(A^c) = 1 - P(A)$
- $A \subset B \Rightarrow P(B \setminus A) = P(B) - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(\bigcup_{n \in \mathbb{N}} A_n) \leq \sum P(A_n)$
- $(A_n)_{n \in \mathbb{N}}$ is increasing $\Rightarrow P(\bigcup_{n \in \mathbb{N}} A_n) = \lim_{n \in \mathbb{N}} P(A_n)$
- $(A_n)_{n \in \mathbb{N}}$ decreasing $\Rightarrow P(\bigcap_{n \in \mathbb{N}} A_n) = \lim_{n \in \mathbb{N}} P(A_n)$

Demo: $(A_n)_{n \in \mathbb{N}}$ is increasing $\Rightarrow P(\bigcup_{n \in \mathbb{N}} A_n) = \lim_{n \rightarrow \infty} P(A_n)$

$$A_0 \subset A_1$$

$$A_1 \cup A_0 = A_0 \sqcup (A_1 \setminus A_0)$$

$$P(\bigcup_{n \in \mathbb{N}} A_n) = P(\bigcup_{n \in \mathbb{N}} (A_n \setminus A_{n-1})) = \sum P(A_n \setminus A_{n-1})$$

$$= \sum P(A_n) - P(A_{n-1}) = \lim P(A_n)$$

($P(A_0) = 0$ since $A_0 = \emptyset$)

Definition: Let $C \in \mathcal{P}(\mathbb{R})$. Generated σ -algebra
 $\sigma(C) = \bigwedge_{\substack{\mathcal{F} \\ \text{\(\sigma\)-algebra} \\ C \subset \mathcal{F}}} \mathcal{F}$ (the smallest σ -algebra that contains C)

If $\Omega \in \mathbb{R}$, there is no "good" proba P , on $\mathcal{P}(\mathbb{R})$
($\rightarrow$ better to have a small set of events countable)

Definition: the Borel σ -algebra is $\sigma(\text{open sets of } \mathbb{R})$
 $= \mathcal{B}(\mathbb{R})$

If $\exists a, b \in \mathbb{R}$, then $(a, b], [a, b),]-\infty, a], \{a\} \in \mathcal{B}(\mathbb{R})$

Theorem: there exists a measure λ (Lebesgue measure) on $\mathcal{B}(\mathbb{R})$ s.t. $\lambda((a, b]) = b - a$ $a \leq b$
 $\hookrightarrow$ not a probability measure

Definition: if f is a positive function

If $\int_{\mathbb{R}} f d\lambda = 1$ then f is a density and

$P((a, b]) = \int_a^b f d\lambda$ define a proba. on $\mathcal{B}(\mathbb{R})$

If $\Omega \in \mathbb{R}^d$, $\mathcal{B}(\mathbb{R}^d) = \underbrace{\mathcal{B}(\mathbb{R}) \times \dots \times \mathcal{B}(\mathbb{R})}_{d \text{ times}}$ (integrate d times)

$$C(A, B) \in \mathbb{R}^2, P_{\mathbb{R}^2}(A, B) = P_{\mathbb{R}}(A) P_{\mathbb{R}}(B)$$

Example: • $U(a, b)$ $f = \frac{1}{b-a}$

• $\lambda > 0$ $\mathcal{E}(\lambda)$

$$f: x \mapsto \lambda e^{-\lambda x} \mathbb{1}_{\mathbb{R}^+}$$

• $\mathcal{N}(\mu, \sigma^2)$

$$\sigma > 0, f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x-\mu^2}{2\sigma^2}}$$

II - Random Variables

Definition: $(E, \mathcal{E}), (F, \mathcal{F})$ σ -algebra

An application $f: E \rightarrow F$ is measurable if $\forall A \in \mathcal{F}, f^{-1}(A) \in \mathcal{E}$

Definition: $(\Omega, \mathcal{F}, P)$ probability space $(E, \mathcal{E})$

A random variable is a measurable application

$$X: (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$$

$A \in \mathcal{E} \Rightarrow X^{-1}(A) \in \mathcal{F}$ so $P(X^{-1}(A))$ events

Notation: $X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\}$

Definition: σ -algebra generated by X
a random variable X

$$\sigma(X) = \sigma(\underbrace{\{X^{-1}(A), A \in \mathcal{E}\}}_{\text{images}})$$

Definition: Distribution of X

$$\forall A \in \mathcal{E}, P_X(A) = P(X^{-1}(A)) = P(X \in A)$$

Notation: $X \sim Y$ means $P_X = P_Y$

$(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$



say mention
de continue

Definition: Repartition function

$$F_X : \begin{cases} \mathbb{R} \rightarrow \mathbb{R} \\ x \mapsto P(X \leq x) = P_X(X^{-1}(-\infty, x]) \end{cases}$$

Theorem: F_X characterizes P_X

Definition: $E(X) = \int_{-\infty}^{\infty} x dP$

$$\text{Var}(X) = E[(X - E(X))^2] = E[X^2] - E[X]^2$$

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

properties:

- Cov is symmetrical and bilinear
- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
- $E[aX + b] = aE[X] + b$
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

Theorem: If $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous, f is a density of P_X

$$E[g(X)] = \int_{\mathbb{R}} g(x) f(x) dx$$

E finite, $E[f(X)] = \sum_{x \in \mathbb{R}} f(x) P(X=x)$

Definition: $L^p = \{f: E \rightarrow \mathbb{R} \mid |f|^p \text{ is integrable}\}$

Markov: if $X \in L^1, \forall a > 0, P(|X| > a) \leq \frac{E(|X|)}{a}$

Demo: $|X| \geq |X| 1_{\{|X| \geq a\}} \geq 0$

$$\begin{aligned} E[|X|] &= \int_{-\infty}^{\infty} |X| dP \geq \int |X| 1_{\{|X| \geq a\}} dP \geq a \int 1_{\{|X| \geq a\}} dP \\ &\geq a P(|X| \geq a) = a P(|X| > a) \end{aligned}$$

Bienayme-Tchebychev: $X \in L^2, a > 0. P(|X - E[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}$

Demo: Markov applied to $\{|X - E[X]|^2 \geq a^2\}$

Consequence: $\text{Var } X = 0 \Rightarrow X = E[X]$ almost surely

Cauchy-Schwarz: $X, Y \in L^2. |E[XY]| \leq \sqrt{E[X^2]E[Y^2]}$

Jensen: $X \in L^1, \varphi$ convex s.t. $\varphi(X) \in L^1$

$$\varphi(E[X]) \leq E[\varphi(X)]$$

INDEPENDANCE:

Definition: • A and B are independent if $P(A \cap B) = P(A)P(B)$

• X, Y random variable if $\forall A \in \mathcal{F}, \forall B \in \mathcal{F}, P(X \in A \cap Y \in B) = P(X \in A)P(Y \in B)$

$X_1, \dots, X_n$ are independent, if $\forall A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$

$$P\left(\bigcap_{i=1}^n X_i \in A_i\right) = \prod_{i=1}^n P(X_i \in A_i)$$

If X_i has density f_i . If $X_1, \dots, X_n$ are independent

$$f_{(X_1, \dots, X_n)}(x_1, \dots, x_n) = f_1(x_1) \times \dots \times f_n(x_n)$$

properties: If X, Y are independent, $E[XY] = E[X]E[Y]$
and $\text{Cov}(X, Y) = 0$

$$P_{X+Y} \sim P_X * P_Y$$

Characteristic function: X a real-valued random variable. $\forall t \in \mathbb{R}, \varphi_X(t) = E(e^{itX})$

If $X: \Omega \rightarrow \mathbb{R}^d, \varphi_X(t) = E(e^{i\langle t, X \rangle}) \forall t \in \mathbb{R}^d$

- properties:
- φ_X characterizes P_X
 - $X \perp Y, \varphi_{X+Y}(t) = \varphi_X(t) \varphi_Y(t)$
 - $X \sim \mathcal{N}(\mu, \sigma^2)$
 $\hookrightarrow \varphi_X(t) = e^{i\mu t - \frac{\sigma^2 t^2}{2}}$
 - $\varphi_{aX+b}(t) = E[e^{i\langle t, b \rangle} e^{i\langle a t, X \rangle}]$
 $= e^{i\langle t, b \rangle} \varphi_X(at) \quad \parallel \begin{matrix} a \in \mathbb{R} \\ b \in \mathbb{R}^d \end{matrix}$

III - Convergence of random variables 15/09/2016

Definition: Let $(X_n)_{n \in \mathbb{N}}$ a sequence of random variables

(X_n) converges almost surely to X

if $P(\{\omega \in \Omega : |X_n(\omega)| \rightarrow X(\omega)\}) = 1$

Notation: $X_n \xrightarrow{\text{a.s.}} X$ almost surely

Definition: (X_n) in $L^p, X \in L^p$

(X_n) converges to X in L^p

if $E[|X_n - X|^p] \rightarrow 0$

$X_n \xrightarrow{L^p} X$

Definition: Let $(X_n), X$ be random variable
(convergence en probabilité)
 (X_n) converges to X in probability
 if $\forall \varepsilon > 0, P(|X_n - X| > \varepsilon) \xrightarrow{n \rightarrow \infty} 0$
 $X_n \xrightarrow{P} X$

Definition: (X_n) converges in law to X
(convergence en loi (la @ faible des convergences))
 if for any function f that is continuous
 and bounded, $E[f(X_n)] \rightarrow E[f(X)]$
 Notation: $X_n \xrightarrow{L} X$

properties:
 $\begin{matrix} a.s. \\ \vdots \\ L^p \end{matrix} \Rightarrow P \Rightarrow L \quad (\text{demo ds TD 1 exo 7})$

properties: if $X_n \xrightarrow{P} X$, there exists a subsequence $(X_{n_k})_{k \in \mathbb{N}}$
 such that $X_{n_k} \xrightarrow{a.s.} X$

Example: $U \sim U(0, 1)$

$$X_n = n \mathbb{1}_{\{U \leq \frac{1}{n}\}}$$

$$\forall \omega \in \Omega, \exists N \text{ s.t. } U(\omega) > \frac{1}{N}$$

$$P(X_n \rightarrow 0) = 1$$

$$X_n \xrightarrow{a.s.} 0$$

$$E[|X_n|] = 1 \quad \text{So no CV in } L^1$$

properties: If $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous, $X_n \xrightarrow{a.s.} X \Rightarrow f(X_n) \xrightarrow{a.s.} f(X)$
 $X_n \xrightarrow{P} X \Rightarrow f(X_n) \xrightarrow{P} f(X)$

properties: $X_n \xrightarrow{L} X$ if and only if for all $x \in \mathbb{R}$ s.t.
 F_X is continuous at x , $F_{X_n}(x) \rightarrow F_X(x)$

If t is discrete, the criteria is:

$$P(X_n = k) \xrightarrow{n \rightarrow \infty} P(X = k) \quad \forall k$$

theorem (Levy): $X_n \xrightarrow{\mathcal{L}} X$ if and only if $\left(\begin{array}{l} \text{prendre la} \\ \text{def de } \mathcal{L} \\ \text{avec flte} \end{array} \right)$
 $\forall t \in \mathbb{R}, \varphi_{X_n}(t) \rightarrow \varphi_X(t)$

Slutsky's lemma: If $X_n \xrightarrow{\mathcal{L}} X, X \xrightarrow{P} c \in \mathbb{R}$
then $(X_n, Y_n) \xrightarrow{\mathcal{L}} (X, c)$

IV - Convergence theorems

$$X_1, \dots, X_n \text{ iid } S_n = \sum_{i=1}^n X_i$$

Law of large number (L^2):

$$\text{If } X_i \in L^2 \text{ and } E[X_i] = m, \frac{S_n}{n} \xrightarrow{L^2} m$$

$$\text{Demo: } e_n = E\left[\left|\frac{S_n}{n} - m\right|^2\right] = \frac{1}{n^2} \text{Var}(S_n)$$

$$E[S_n] = nm$$

$$\text{Var}(S_n) = n \text{Var}(X_1) \quad (\text{finite since } L_2)$$

$$e_n = \frac{\text{Var}(X_1)}{n} \rightarrow 0$$

Strong law large number: $X_i \in L^1, \frac{S_n}{n} \xrightarrow{\text{a.s.}} m$

Central Limit Theorem: $X_1, \dots, X_n$ iid with $E[X_i] = m$
and $\text{Var}(X_i) = \sigma^2$
 $\frac{S_n - nm}{\sigma \sqrt{n}} \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1)$

V - Concentration

(L) Results with finite sample size

Hoeffding's inequality: $\mathcal{X}$

$X_1, \dots, X_n$ iid, $E[X_i] = m$ s.t. $\forall i, a \leq X_i \leq b$
a.s. $\forall \varepsilon > 0, P(|\frac{S_n}{n} - m| > \varepsilon) \leq 2e^{-\frac{2n\varepsilon^2}{(b-a)^2}}$

VI - Random vectors

Definition: $X = (X_1, \dots, X_d)^T \in \mathbb{R}^d$ (random variable on $\mathcal{B}(\mathbb{R}^d)$)

$$E(X) = (E[X_1], \dots, E[X_d])$$

$$\text{Covariance matrix: } \Sigma_X = E[(X - E(X))(X - E(X))^T]$$

$$\Sigma_X \in \mathcal{M}_d(\mathbb{R})$$

$$\Sigma_{X(i,j)} = \text{Cov}(X_i, X_j)$$

property: $\bullet E[AX + b] = AE[X] + b$
 $A \in \mathcal{M}_d(\mathbb{R}) \leftarrow \mathbb{R}^d$

$$\bullet \Sigma_{AX+b} = A \Sigma_X A^T$$

$\bullet \Sigma_X$ is symmetric, positive semi-definitive

$$\forall x \in \mathbb{R}^d, x^T \Sigma_X x = \text{Var}(x^T X) \geq 0$$

Definition: A random vector X is a Gaussian vector if $\forall a \in \mathbb{R}^d, a^T X \sim \mathcal{N}(m, \sigma^2)$ for some

property: If X is Gaussian with $E[X] = \underset{\text{vector}}{m}, \Sigma_X$ $m, \sigma \in \mathbb{R}$
 $E[e^{it^T X}] = \varphi_X(t) = e^{i t^T m - \frac{1}{2} t^T \Sigma_X t} \quad (X \sim \mathcal{N}(m, \Sigma_X))$

property: If X is Gaussian, it has a density
 $\sim N(m, \Sigma_X)$ with Σ_X invertible

$$f(x) = \frac{1}{(\sqrt{2\pi})^d |\Sigma_X|^{1/2}} e^{-\frac{(x-m)^T \Sigma_X^{-1} (x-m)}{2}}$$

Theorem: If X is Gaussian, $X_1, \dots, X_n$ are independent
 $\Leftrightarrow \Sigma_X$ is diagonal

Demo: If $\Sigma_X = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \ddots & 0 \\ 0 & \ddots & \sigma_d^2 \end{pmatrix}$

$$\forall t, \varphi_X(t) = e^{i \langle t, m \rangle - \frac{1}{2} \sum_{j=1}^d t_j^2 \sigma_j^2}$$

$$= \prod_{j=1}^d e^{i t_j m_j - \frac{1}{2} t_j^2 \sigma_j^2}$$

$(X_1, \dots, X_n)$ ind. $\Leftrightarrow$
 var demo $\varphi(t) = \prod \varphi_j(t_j)$

$$= \prod_j \varphi_{X_j}(t_j) \Rightarrow X_1, \dots, X_n \text{ ind}$$

VII - Conditioning (TD exc 15-16)

If $P(B) > 0$, $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$P_B = P(\cdot | B)$ is a probability measure

$$\hookrightarrow E_B[X] = \int X dP_B$$

$$\Omega = \bigsqcup_{i=1}^n A_i$$

$$P(B) = \sum_{i=1}^n P(B \cap A_i) = \sum_{i=1}^n P(B|A_i) P(A_i)$$

$$\text{Bayes: } P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

E is finite

X, Y random variable

If $P(Y=y) > 0$

$$P(X \in A | Y=y) = \frac{P(X \in A, Y=y)}{P(Y=y)}$$

$$E[X | Y=y] = \sum k P(X=k | Y=y)$$

Consider the function $g: \begin{cases} E \rightarrow \mathbb{R} \\ y \mapsto E[X | Y=y] \end{cases}$
↳ this is a random variable variable called $E[X|Y]$

VIII - Markov chains

E finite

definition: A Markov chain is a process $(X_n)_{n \in \mathbb{N}}$ such that $\forall n, \forall x_0, x_1, \dots, x_n, y \in E,$

$$P(X_{n+1}=y | X_0=x_0, \dots, X_n=x_n)$$

$$= P(X_{n+1}=y | X_n=x_n)$$

transition matrix: $Q(x, y) = P(X_1=y | X_0=x)$

property: $\forall n \geq 1, P(X_n=x | X_0=y) = Q^n(y, x)$

If $\mu_0 = (p_0, \dots, p_d)$ is the distribution of probabilities at step 0.
↳ list of probas

then $\mu_n = \mu_0 Q^n$

Definition: A distribution π is stationary if $\pi = \pi Q$

↳ $\mu_0 = \pi, \mu_n = \pi \forall n$

Definition: A Markov chain is irreducible if each state can be reached from any other state (maybe in several steps)

Definition: it is aperiodic if there exists no periodic return to any state

Theorem: a finite irreducible aperiodic Markov chain has a unique stationary distribution π .

And $Q^n(x, \cdot) \xrightarrow[n \rightarrow \infty]{} \pi$ for each initial state x
(n.c.m.c)
 $\uparrow$
any state y